

28/08/2026

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- EXERCISE ON DERIVATION OF STATE SPACE MODEL OF A DYNAMICAL SYSTEM.
- GENERAL MPC FORMULATION $\rightarrow$ FOR THEORETICAL STUDIES
- DUAL MODE CONTROL, VALUE FUNCTIONS.

MODELING - STATE SPACE

NONLINEAR

$$\begin{cases} \dot{x}(t) = f(x(t), u(t)) \\ y(t) = g(x(t), u(t)) \end{cases} \quad \begin{array}{l} \text{STATE} \\ \text{EQ.} \\ \\ \text{OUTPUT} \\ \text{EQ.} \end{array}$$

LINEAR SYSTEM

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) + Du(t) \end{cases}$$

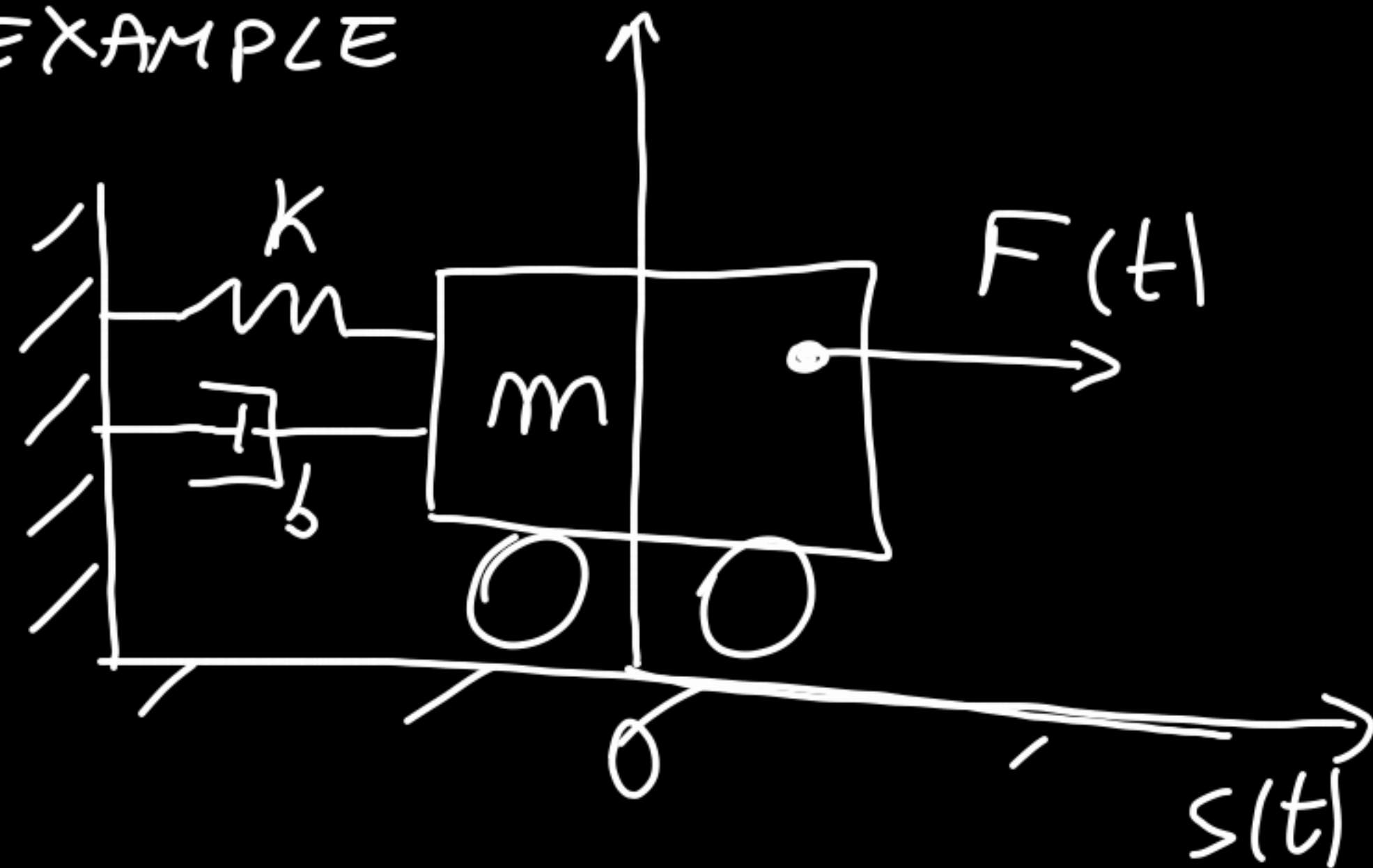


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$$\begin{cases} x_{k+1} = f(x_k, u_k) \\ y_k = g(x_k, u_k) \end{cases}$$

$$\begin{cases} x_{k+1} = Ax_k + Bu_k \\ y_k = Cx_k + Du_k \end{cases}$$

EXAMPLE



CONTROL GOAL:

CONTROL HORIZONTAL
DISPLACEMENT OF MASS, $s(t)$,
BY ACTING ON $F(t)$

NEWTON LAW: $m \cdot a = \sum F_e$

Q : ACCELERATION $Q = \frac{d}{dt} v$

v : VELOCITY $v = \frac{d}{dt} s$

③

$$m \overset{\frac{d}{dt} v}{a} = F(t) - k s(t) - b \overset{\frac{d}{dt} s}{v}(t)$$

④
STEP 1 - FIND THE
RELEVANT EQUATIONS

$$\begin{cases} \dot{x} = ? \\ y = ? \end{cases}$$

STEP 2: IDENTIFY THE STATE
VARIABLES. LOOK FOR THE
VARIABLES WE DERIVE.

$$\boxed{x_1 = s ; x_2 = v}$$

$$\left(\text{IDENTIFY } \underline{u, y} \begin{cases} u = F \\ y = s \end{cases} \right)$$

STEP 3 : REWRITE THE EQUATIONS USING GIVEN (5)

DEFINITIONS FOR x, u, y .

$\dot{x} = ?$

$$\begin{cases} \dot{x}_1 = \frac{d}{dt} s = v \stackrel{\text{DEF}}{=} x_2 \end{cases}$$

$$\begin{cases} \dot{x}_2 = \frac{d}{dt} v = a = \frac{1}{m} F(t) - \underbrace{\frac{k}{m} s(t) - \frac{b}{m} v(t)}_{\text{NEWTON LAW}} = \end{cases}$$

$$= -\frac{k}{m} x_1 - \frac{b}{m} x_2 + \frac{1}{m} u(t)$$

$$\begin{cases} \dot{x}_1 = 0x_1 + 1x_2 + 0u(t) \\ \dot{x}_2 = -\frac{k}{m}x_1 - \frac{b}{m}x_2 + \frac{1}{m}u(t) \\ y = 1x_1 + 0x_2 + 0u \end{cases}$$

SCALAR
FORM

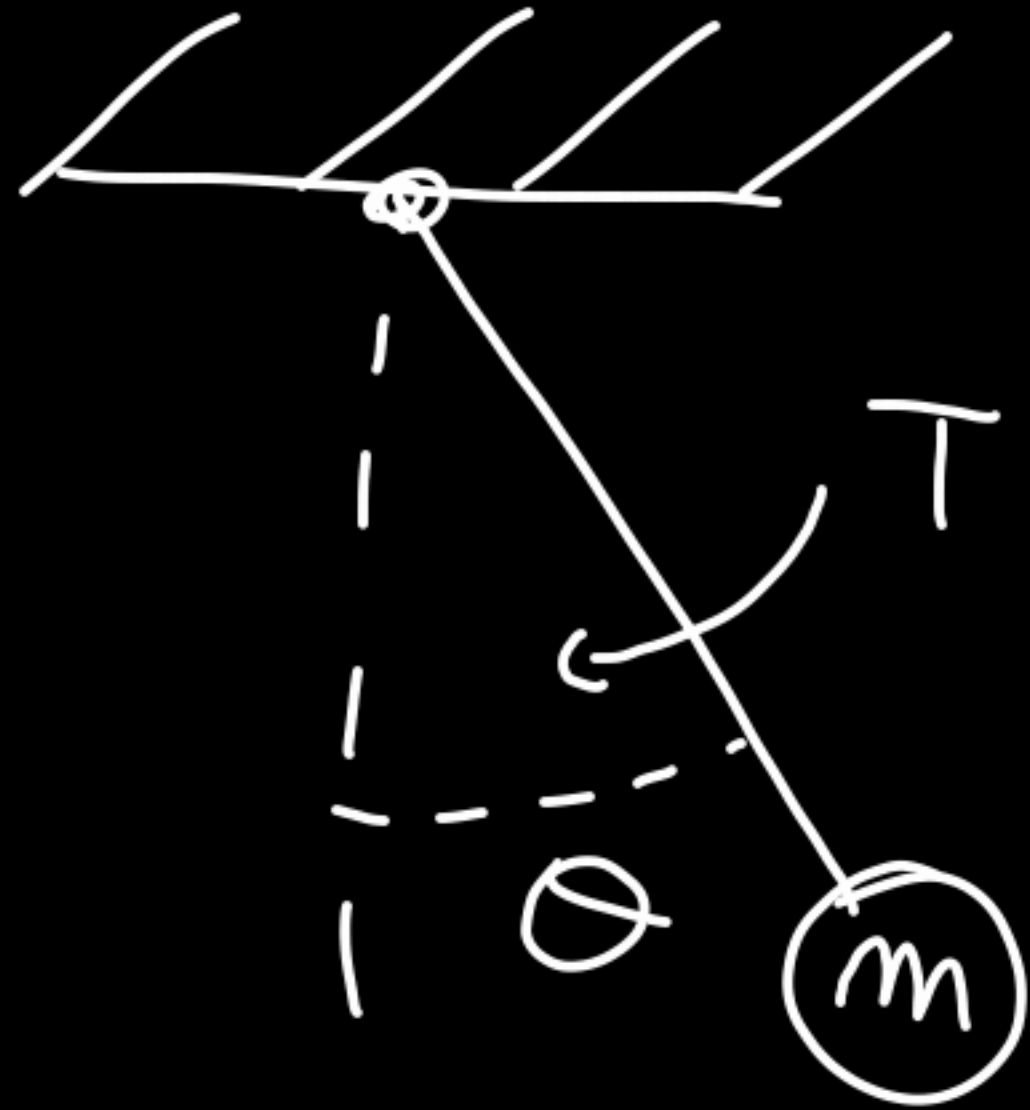
ONE INPUT, ONE OUTPUT,
TWO STATES.

LINEAR SYSTEM

$$\begin{cases} \dot{x} = \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{pmatrix} x + \begin{pmatrix} 0 \\ \frac{1}{m} \end{pmatrix} u \\ y = \begin{pmatrix} 1 & 0 \end{pmatrix} x + \underbrace{0}_{(D)} u \end{cases}$$

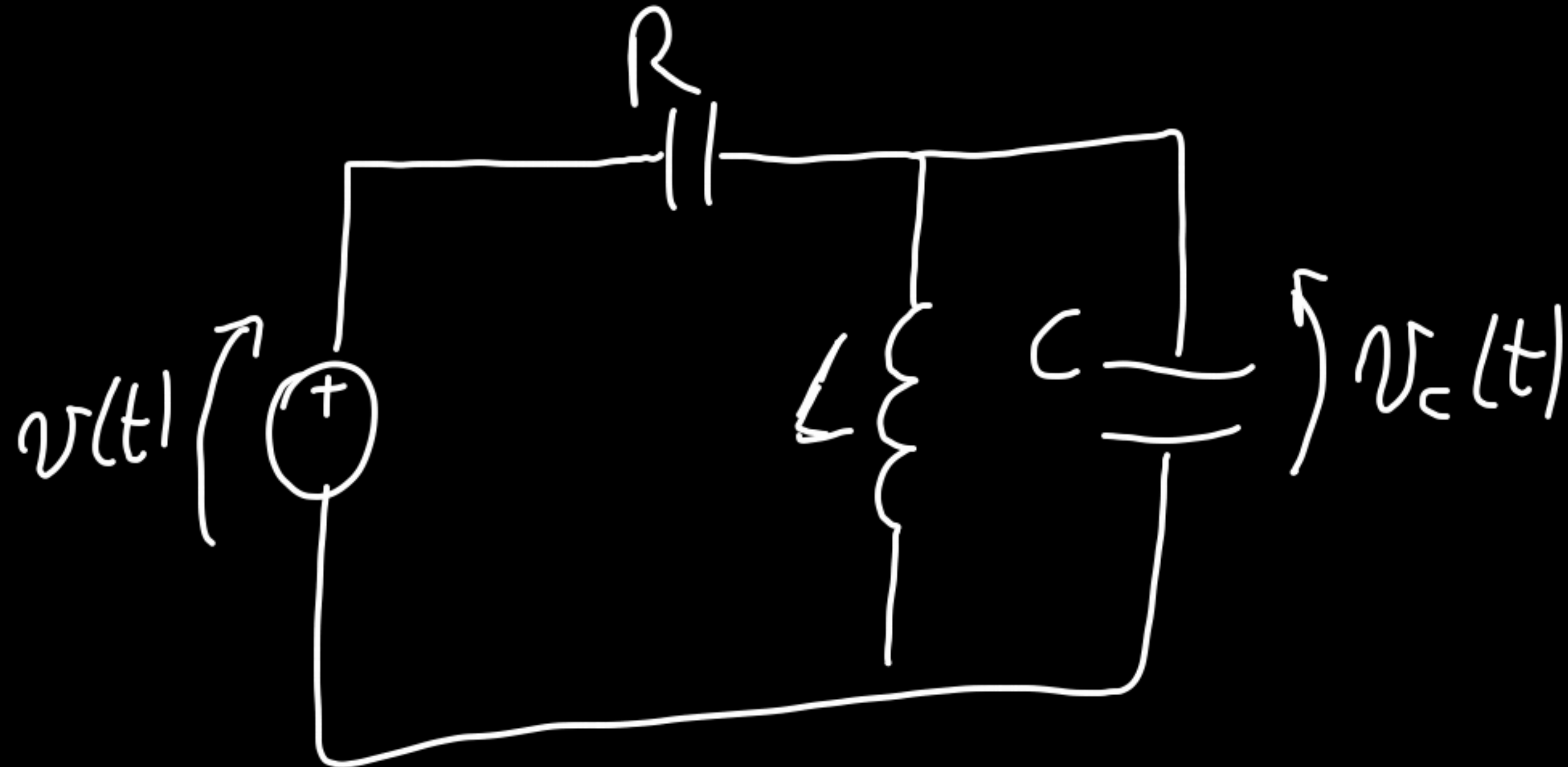
(A) (B) (C) (D)

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases}$$



GOAL: CONTROL θ BY ACTING
ON TORQUE $T(t)$.

NONLINEAR MODEL.



LINEAR.

GENERAL FORMULATION OF MPC PROBLEM:

$\forall$ TIME k :

① MEASURE CURRENT STATE x_k AND ANY OTHER RELEVANT INFO;

② BUILD AND SOLVE MPC PROBLEM $\mathcal{P}_k$ $\rightarrow$ WE NOW WRITE A GENERAL FORMULATION OF THIS.

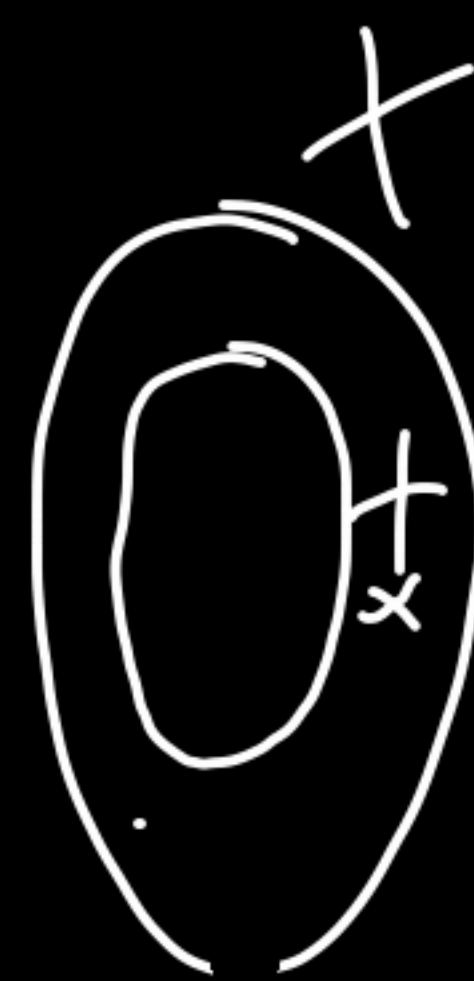
③ APPLY FIRST SOLUTION.

p

GENERAL FORMULATION FOR P_k :

$$P_k: \min \left\{ V = \sum_{i=0}^{N-1} \overset{\text{STAGE COST}}{\ell(x_{i|k}, u_{i|k})} + \overset{\text{TERMINAL COST}}{V_f(x_{N|k})} \right\}$$

SUBJECT TO:



$$\begin{cases} x_{i+1|k} = f(x_{i|k}, u_{i|k}), i=0, 1, \dots, N-1 & \text{DYNAMICS OF SYSTEM.} \\ x_{0|k} = x_k & \text{(INITIAL CONDITION)} \end{cases}$$

$$\begin{cases} x_{i|k} \in X, i=0, 1, \dots, N & \text{ADMISSIBLE STATES} \\ u_{i|k} \in U, i=0, 1, \dots, N-1 & \text{ADMISSIBLE CONTROLS} \\ x_{N|k} \in X_+, \text{ TERMINAL CONSTRAINT} \end{cases}$$

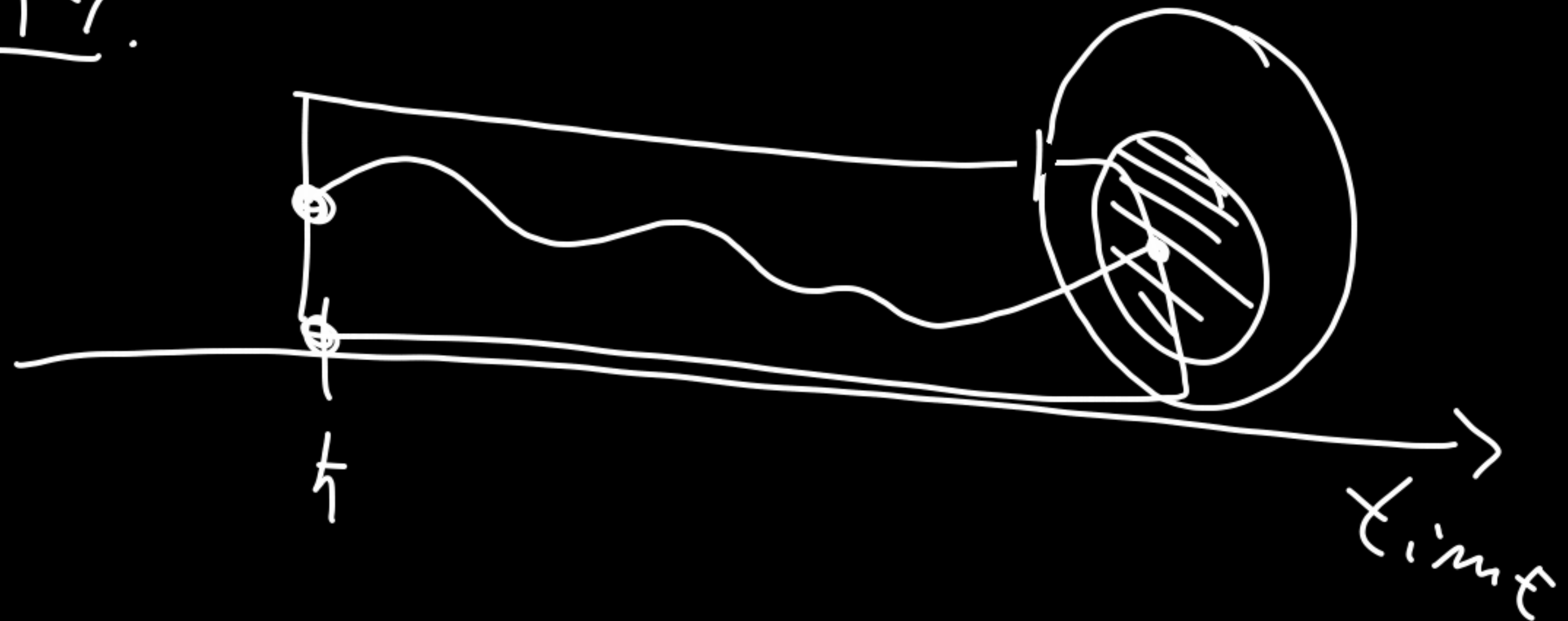
MPC "INGREDIENTS"

- N - CONTROL HORIZON. IF WE INCREASE IT WE HAVE BETTER PERFORMANCE BUT LONGER COMP. TIMES.
- ℓ - STAGE COST - IT IS CHOSEN TO REFLECT THE CONTROL GOAL.
- V_f - TERMINAL COST - IT HAS STRONG IMPACT ON MPC PERFORMANCE (OPTIMALITY GAP).
- MODEL - COMES FROM APPLICATION

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• X, U ADMISSIBLE STATES AND CONTROLS - THEY COME FROM THE APPLICATION.

• X_f TERMINAL SET. VERY IMPORTANT TO

SELECT IT PROPERLY TO ENSURE RECURSIVE
FEASIBILITY.



EXAMPLES ON ℓ .

$$\ell(x_{ik}, u_{ik}) = \underline{a} (x_{ik} - x^{ref})^2 + \underline{b} u_{ik}^2 \quad \text{QUADRATIC FUNCTION}$$

IN GENERAL, FOR REGULATION PROBLEMS ℓ
IS QUADRATIC IN THE STATE AND CONTROL

$$\ell(x_{ik}, u_{ik}) = x^T Q x + u^T R u$$

$$\begin{aligned} x &\in \mathbb{R}^n \\ u &\in \mathbb{R}^p \\ Q &\in \mathbb{R}^{n \times n} \\ R &\in \mathbb{R}^{p \times p} \end{aligned}$$

ASSUMPTIONS ON Q, R

$$(x^{\text{ref}} = 0)$$

$\square Q > 0$ Q POSITIVE-DEFINITE:

- $x^T Q x > 0, \forall x \neq 0$

$\square R > 0$ $\parallel$

SOMETIMES

$Q \geq 0$ SEMI-POSITIVE DEFINITE

$$x^T Q x \geq 0 \quad \forall x.$$

EXAMPLE

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} s \\ v \end{pmatrix}$$

$$\begin{pmatrix} * \\ 0 \end{pmatrix} \neq 0$$

GOAL: REGULATE

$x \rightarrow 0$

$s \rightarrow 0$

$v \rightarrow 0$

$$Q(x_{ilk}, u_{ilk}) = (x_1, x_2) \begin{pmatrix} \overbrace{0 \quad 0}^I \\ 0 \quad 10 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + 1 \cdot u_{ilk}^2$$

$Q \geq 0$

$$P_k : \min_{\{u_{0|k}, u_{1|k}, \dots, u_{N-1|k}\}} \left\{ V_k(x_k, \underline{\quad}) \right\} = \sum_{i=0}^{N-1} \rho(x_{i|k}, u_{i|k}) + V_f(x_{N|k})$$

s.t.:

$u_{0|k}^*$ IS A
FUNCTION
OF x_k .

$$\begin{cases} x_{i+1|k} = f(x_{i|k}, u_{i|k}), i=0, 1, \dots, N-1 \\ x_{0|k} = \underline{x_k} \end{cases}$$

$$\begin{cases} x_{i|k} \in X, i=0, 1, \dots, N \\ u_{i|k} \in U, i=0, 1, \dots, N-1 \end{cases}$$

$$x_{N|k} \in X_f$$

$$\mathcal{U}_k \triangleq \{u_{0|k}, u_{1|k}, \dots, u_{N-1|k}\}$$

CONTROL SEQUENCE
AT TIME k

$$\bar{x}_k \triangleq \{x_{0|k}, x_{1|k}, \dots, x_{N|k}\}$$

PREDICTED STATE
SEQ.

MPC FLOW:

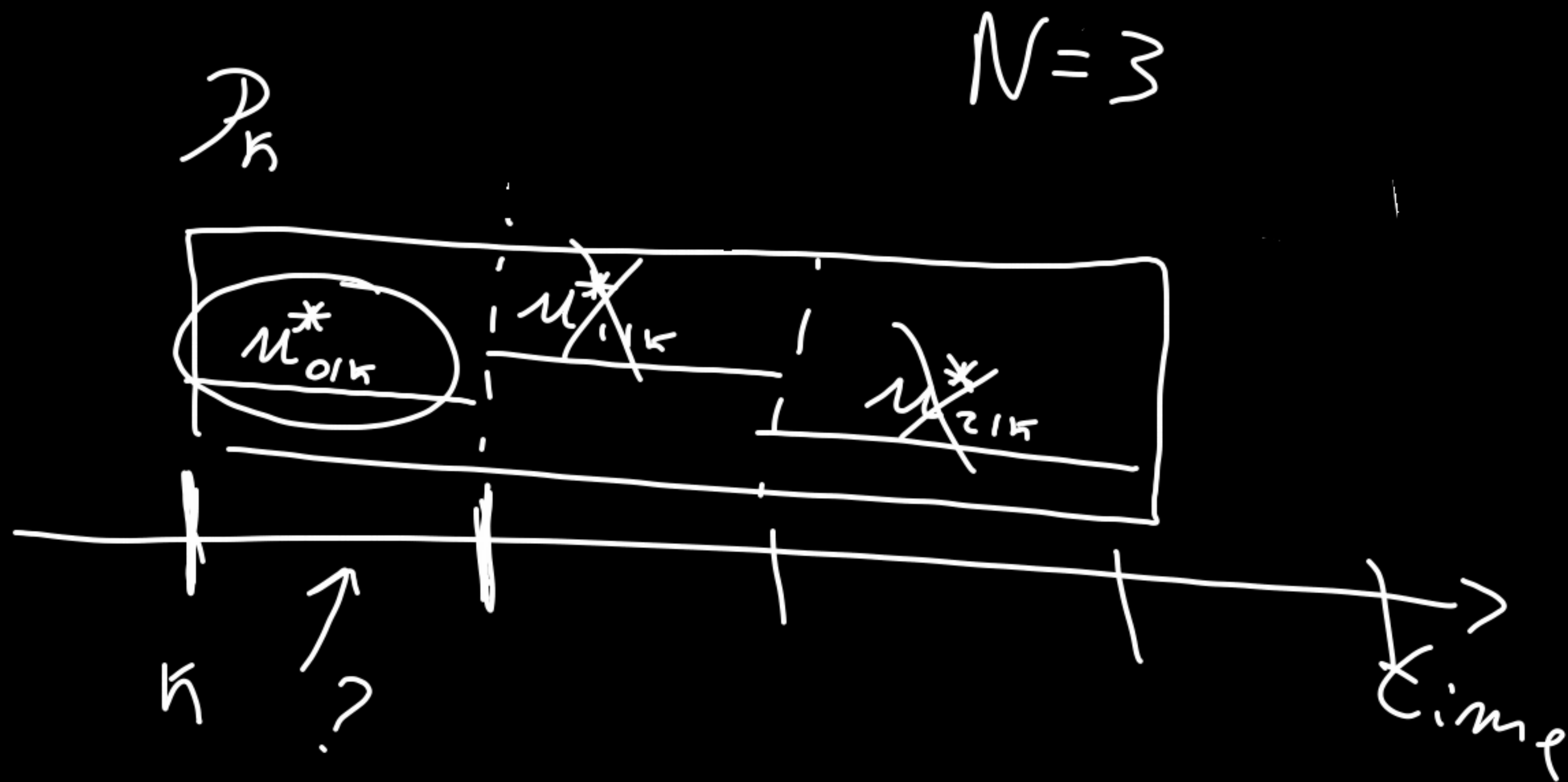
① MEASURE x_k

② BUILD AND SOLVE $\mathcal{P}_k$

③ APPLY $u_{0|k}^*$

OPTIMAL CONTROL SEQUENCE

$$\mathcal{U}_k^* \triangleq \{u_{0|k}^*, u_{1|k}^*, \dots, u_{N-1|k}^*\}$$



$$\mu_{0|k}^* = k($$

MPC IS A FEEDBACK CONTROL TECH. $\rightarrow$ CONTROL
IS A FUNCTION OF THE CURRENT STATE

$$u_{0|k}^* = K(x_{0|k})$$

INFINITE-HORIZON / CONSTRAINED CONTROL (∞ -HORIZON CONSTR.)

"IDEAL PROBLEM" WE WOULD LIKE TO SOLVE

(k, x_k) CURRENT TIME - STATE

$$\min_{u_k, u_{k+1}, u_{k+2}, \dots} \left\{ V^\infty = \sum_{i=0}^{\infty} \ell(x_{k+i}, u_{k+i}) \right\}$$

x_k KNOWN SUBJECT TO:

$$\begin{aligned} &\rightarrow x_{k+i} \in X, i=0, 1, 2, \dots \\ &\rightarrow u_{k+i} \in U, i=0, 1, 2, \dots \\ &x_{k+i+1} = f(x_{k+i}, u_{k+i}), i=0, 1, 2, \dots \end{aligned}$$

FIRST SIMPLIFICATION. FROM GENERAL ∞ -HORIZON^{CONSTRAINED} CONTROL PROBLEMS,
TO ∞ -HORIZON LINEAR-QUADRATIC CONSTRAINED
REGULATION PROBLEMS. (∞ -HORIZON CONSTRAINED LQR
PROBLEMS).